RÉPUBLIQUE FRANÇAISE
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Concours d'entrée 2026

Voie « Orient »

Concours externe

4ème épreuve d'admissibilité

Une épreuve consistant en des questions à réponses courtes, en anglais, sur les enjeux globaux, notamment économiques, internationaux, du développement et des négociations climatiques internationales.

Durée : 3 heures – coefficient 2.

L'épreuve de questions à réponses courtes consiste à traiter de manière concise des sujets transversaux portant sur les enjeux mondiaux du développement en anglais. Cette épreuve doit être abordée dans une perspective interdisciplinaire, en considérant la langue anglaise comme véhicule de transmission d'un argumentaire analytique pouvant déboucher sur des conclusions personnelles. Elle vise à vérifier que les candidats disposent d'une bonne maîtrise écrite de la langue anglaise et sont en capacité à mobiliser leurs connaissances avec pertinence pour les exprimer en anglais.

L'épreuve vise à évaluer la capacité à écrire en anglais des textes clairs et détaillés sur les questions relatives aux enjeux mondiaux en faisant la synthèse et l'évaluation d'informations et d'arguments empruntés au dossier proposé. Outre la maîtrise des connaissances, seront valorisées la capacité à exposer en anglais, clairement et de manière bien structurée, les réponses attendues et l'aptitude à exposer puis prouver son point de vue assez longuement à l'aide d'arguments secondaires, de justifications et d'exemples pertinents sur les enjeux mondiaux.

Chaque question posée peut être accompagnée d'un ou de plusieurs documents (notamment articles, graphiques, tableaux statistiques) à expliquer et commenter. Leur lecture rigoureuse et leur analyse doivent être mises au service de la réflexion personnelle des candidats en vue de traiter la question posée. Un même document peut servir de support à plusieurs questions.

Sujet

Question 1 : To what extent is generative artificial intelligence a boon for the economy? (7 points)

Question 2 : Is subsea infrastructure the new warfare? (7 points)

Question 3 : How can vaccine scepticism be addressed? (6 points)

Dossier

N°	Documents joints	Pages
1	"Beyond the Action Plan: Towards a Holistic Strategy for a Competitive and Secure Subsea Infrastructure in Europe", discussion paper by Varg Folkman, Mihai Sebastien Chihaia and Ian Hernandez, 20 October 2025, *European Policy Centre, www.epc.eu* (extraits)	5 et 6
2	"AI Sovereignty and the Geopolitics of Submarine Cables", by Olivier Chatain and Jeremy Ghez, Spring 2025, *Knowledge@HEC, www.hec.edu*	7 et 8
3	"Redundancy, Resiliency, and Repair - Securing Subsea Cable Infrastructure", by Erin L. Murphy, November 2025, *Center for Strategic and International studies, www.csis.org* (extraits)	9 à 11
4	"Is generative AI a general-purpose technology? Implications for productivity and policy", by Flavio Calvino, Daniel Haerle, Sarah Liu, June 2025, *OECD Artificial Intelligence Papers No. 40, www.oecd.org* (extrait)	12 et 13
5	"Canada fought measles and measles won; virus now endemic after 1998 elimination", by Beth Mole, 10 November 2025, *Ars Technica, www.arstechnica.com*	14 et 15
6	"PAHO calls for regional action as the Americas lose measles elimination status", 10 November 2025, *Pan American Health Organisation, www.paho.org*	16 et 17
7	"Kennedy Says He Told C.D.C. to Change Website's Language on Autism and Vaccines", 21 November 2025, by Sheryl Gay Stolberg, *The New York Times, www.nytimes.com* (extrait)	18 à 20

Liste des sigles :

- AI : Artificial Intelligence
- C.D.C. : Centers for Disease Control and Prevention
- CSIS : Center for Strategic and International Studies
- EPC : European Policy Center
- EU : European Union
- GPT : General-Purpose Technology
- HEC Paris : Ecole des Hautes Etudes Commerciales de Paris
- ICPC : International Cable Protection Committee
- IMI : Invention of a method of invention
- ISA : International Space Agency
- IT : Information Technology
- ITU : International Telecommunication Union
- MMR : Measles, mumps and rubella
- NATO : North Atlantic Treaty Organization
- OECD : Organisation for economic co-operation and development
- PAHO : Pan American Health Organisation
- PCR : Polymerase Chain reaction
- PHAC : Public Health Agency of Canada
- RVC : Regional Verification Commission for Measles and Rubella Elimination
- UK : United Kingdom
- UN : United Nations
- UNCLOS : United Nations Convention on the Law of Sea
- US : United States of America

Document 1

"Beyond the Action Plan: Towards a Holistic Strategy for a Competitive and Secure Subsea Infrastructure in Europe", discussion paper by Varg Folkman, Mihai Sebastien Chihaia and Ian Hernandez, 20 October 2025, *European Policy Centre,*

www.epc.eu (extraits)

Executive summary

Once an invisible backbone of global connectivity, subsea cables are quickly becoming a strategic concern. As demand for internet services rises, investment in new fibre-optic infrastructure is accelerating – and with it, geopolitical attention. The seabed has emerged as a new frontier of hybrid warfare among Europe, Russia and other actors.

Ownership dynamics in the sector are also shifting. Just four US cloud providers now control 71% of global subsea fibre capacity, overtaking traditional telecom carriers and driving most new projects. Europe lags behind in new seabed infrastructure, as its carriers struggle to keep up with their counterparts across the Atlantic. Although the EU funds strategic projects, its investment is dwarfed by cloud hyperscalers.

Awareness of cable networks' strategic value is growing. Concerns over vendors from unreliable countries in subsea and other critical infrastructure and vulnerabilities to hybrid attacks are prompting the EU and NATO to coordinate more closely. But more needs to be done, particularly in public-private cooperation: Europe's capacity to repair damaged cables is limited, and private repair and maintenance agreements are not scaled to respond to coordinated seabed attacks.

Furthermore, there are few global standards ruling the subsea industry. Few laws cover modern issues like acts of hybrid warfare, and international governing bodies remain private initiatives.

To respond, the EU must follow and go beyond the European Commission's 2024 Action Plan on Submarine Cable Security. A comprehensive strategy should accelerate cross-border mapping and risk assessment. This includes identifying vulnerabilities and strengthening repair and resilience capabilities across jurisdictions. […]

Subsea battleground

Subsea infrastructure suffers frequent disruption – about 200 incidents each year, according to the International Cable Protection Committee. While most faults and shunts are caused by natural events such as seismic activity, fishing gear, anchors, or weather phenomena, the number of deliberate attacks has grown. This trend has heightened awareness of the seabed as a new frontier of hybrid warfare.

Recent years have seen a series of high-profile incidents. In 2022, telecom cables to Svalbard cable were disrupted, followed by damage to the Balticconnector gas pipeline and two telecommunications cables between Finland and Estonia in 2023. In 2024, a telecommunications cable linking Sweden and Lithuania was cut, as was another from Finland to Germany. That December, the EstLink power cable and four additional telecommunications lines were damaged.

The scale of the subsea network has grown dramatically over the last decade – from roughly one million kilometres in 2014 to 1,7 million in 2025. This expansion has created a complex tapestry demanding engagement by governments, armed forces, international organisations and private operators.

The nature of threats has also evolved. Incidents now range from service and communications disruptions to hostile actors testing military response capabilities – how many simultaneous cuts can Europe handle? Many intentional incidents employ grey zone tactics: for instance, commercial ships

sailing under flags of third countries or linked to ghost companies, making attribution and defence difficult.

Vulnerabilities are not confined to the seabed. Cable landing stations – where undersea cables connect to terrestrial networks – are often accessible by land and lightly protected. Any strategy to secure Europe's subsea infrastructure must consider these onshore nodes.

Table 1. Timeline of incidents.

2022	Nord Stream pipeline sabotage
2022	Svalbard telecom cables disrupted
2023	Balticconnector gas pipeline and two telecommunications cables damaged
2024	Red Sea cables damaged
2024	Two submarine telecommunication cables, the BCS East-West Interlink and C-Lion1 fibre-optic cables, disrupted in the Baltic Sea
2024	EstLink power cable and four telecommunications cables damaged
September 2025	Multiple cable breaks in the Red Sea

Document 2

"AI Sovereignty and the Geopolitics of Submarine Cables", by Olivier Chatain and Jeremy Ghez, Spring 2025, *Knowledge@HEC,*

www.hec.edu

As the AI arms race ramps up tensions across the globe, the "geopolitical honeymoon" of the early days of the Internet is over, according to Jeremy Ghez and Olivier Chatain. In their ongoing research, funded by the HEC Foundation, the two HEC professors warn that this AI-driven "industrial revolution" could end in a "very messy separation" between the US and China, centered on submarine cables. As the "great divorce" gathers speed, how can businesses adapt and leverage this new infrastructure?

From Shared Networks to Sovereign Ambitions

They're no thicker than a garden hose – albeit a hose encased in galvanized armor and clad in a polyethylene jacket. And yet these bundles of optical fibers buried beneath the ocean floor power up to 95% of all the data that zips around the globe. In all, there are 700,000 nautical miles of undersea cables. Even hidden from sight, there's no escaping Google and Meta (and, to a much lesser extent, Microsoft and Amazon), the heaviest investors in submarine cables. As for Elon Musk, his Starlink satellites simply can't match the massive amounts of data that cable can handle...

But how do these fiber optic cables stand up to the geopolitics of our planet? For ours is a tumultuous age. The trade showdown between the US and China, the ongoing wars in Sudan and Ukraine, the Israeli invasion of Gaza and the revolution in Syria are all stoking international divisions. And, looming over all this is the shadow of climate change.

These geopolitical frictions are shaping the AI revolution. "In the past, technology united us, and interdependence was a source of stability", explains Jeremy Ghez, the academic director of the HEC Center for Geopolitics, "but now these geopolitical rivalries have become technological rivalries". Governments worldwide are being driven by what Olivier Chatain, professor of competitive strategy at HEC, calls "political calculus". They are all bidding to be masters of their digital destiny. AI sovereignty is the name of this new game. Some entities – the US and the EU, for example – have already rolled out their own AI regulations and legislation to ringfence their national interests. If the trend is for control, the potential upshot is "global fragmentation, a patchwork of markets with little consistency, concedes Ghez.

Submarine Kingpins of the Global Economy

But let's come back to our underwater cables. Alongside profit-driven companies and the global telecom carriers, state actors rely on undersea networks for their communications. Although governments rarely own the infrastructure, they sometimes control routes indirectly via state-owned telecom operators. "In fact, the competition between rival states is fiercer than the competition between rival firms," claims Chatain.

Wrapped up in these undersea cables is a startling paradox: this "deep tech" is highly vulnerable. Sharks have been known to take a bite, ship anchors are a menace, and earthquakes pose a problem. Redundancy and swift repairs normally mitigate the impact of a single cut, but a series of incidents may together create prolonged disruptions as in South Africa in March 2024. More significantly, there is the threat of covert attacks stemming from the geopolitical tensions among Washington, Beijing, Russia and the EU. A series of cable cuts in the Baltic and Red Seas in late 2024 and around Taiwan in early 2025 raised eyebrows and have been investigated as instances of hostile actions below the threshold of war.

Governments monitor where new cables are installed. In 2020, the White House vetoed a planned subsea data cable between the US and Asia owned by Google and Meta and subsequently imposed a

re-routing away from China out of concern for communication interceptions. Google is now developing networks in the Pacific and Indian ocean in an apparent bid to avoid the South China Sea. In late 2024, France took the decision to nationalize a strategic submarine network manufacturer and installer (Alcatel Submarine Networks) to prevent this "national security asset" from falling into the wrong hands.

Double-edged Sword Beneath the Waves

This technological revolution is generating countless opportunities for a myriad of actors. "But not necessarily good actors," points out Chatain. What would happen if cables are sabotaged in a coordinated way?" he asks. There would be little – or no – connectivity: national security communications would be compromised, financial transactions would grind to a halt, and critical services would hang in the balance. Interestingly, actors like Google, by investing massively in cable networks in out-of-the-way locations build redundancy and make the overall network more robust.

Both Chatain and Ghez agree it's time to start thinking outside the (business) box. This new AI landscape is dominated not just by inter-firm rivalry, but also by rivalry – and conflict – between nations. As for civil society, it needs to ask itself if it wants to be a bystander, a pawn or an actor.

Article based on an interview with Olivier Chatain, professor of competitive strategy at HEC, and Jeremy Ghez, academic director of the HEC Center for Geopolitics. This ongoing research is funded by the HEC Foundation.

Document 3

"Redundancy, Resiliency, and Repair - Securing Subsea Cable Infrastructure", by Erin L. Murphy, November 2025, *Center for Strategic and International studies*,

www.csis.org (extraits)

Executive Summary

Subsea fiber-optic cables are the world's primary conduit for data, carrying 99 percent of data internationally, making them essential to the modern digital world and indispensable to both national and economic security. Subsea cable infrastructure impacts nearly all aspects of daily life by providing access to the internet and delivering the data that underlies critical systems such as e-commerce and financial networks, communications, and telehealth and e-education. It is also a key part of the foundation for seismic, cutting-edge technologies such as AI, cloud computing, and quantum computing. Although the first subsea cable was laid between Dover and Calais nearly 175 years ago, this critical type of infrastructure shows no signs of ebbing in importance.

Today, more than 600 cables span 1.5 million kilometers globally and connect to more than 1,600 landing stations. As a key component determining access to data and digital systems, this globe-spanning technology also carries significant strategic influence—access to data is a key currency of the modern era and is increasingly being counted among the critical components of geopolitical power, particularly as AI systems proliferate, including into military applications. As geopolitical tensions continue to rise and digital demands grow, the security, resiliency, and maintenance of cables have become urgent global priorities.

[…]

Repair, Redundancy, and Resiliency

The ecosystem to build, lay, and repair cables is small, consisting of four major manufacturers, and the market for cables today is dominated by a handful of hyperscalers—major tech companies that are driving the expansion of subsea fiber-optic cable networks. The four firms that dominate cable production include SubCom (United States), NEC (Japan), Alcatel Submarine Networks (France), and HMN Tech (China). Hyperscalers—including Google, Meta, Microsoft, and Amazon Web Services (AWS)—are the modern tech giants that provide cloud computing, digital infrastructure, and data processing and storage. These companies are increasingly investing, owning, and operating their own subsea fiber-optic cables or joining consortiums with global telecommunications companies or tech leaders. Collectively, manufacturers and hyperscalers are dependent on access to specialized ships, submersible vehicles that can lay and bury the cables, and the skilled crews and technicians necessary to manufacture, lay, repair, and test the cables. These stakeholders must navigate a complex planning, permitting, and financing environment to support new projects and keep existing cables operational.

Outside of new cable projects, the owners and operators of the cable must keep up with infrastructure maintenance and annual breakages. The majority of cable damage is caused by dropped anchors from commercial and fishing vessels that scrape across the ocean floor. Natural disasters, including earthquakes, undersea volcanoes, and storms, can also damage cables and cable landing stations onshore—threats that are likely to increase as climate change intensifies. Less likely, but of considerable concern to policymakers, is intentional sabotage or interference by state and non-state actors. […]

Governments and international organizations provide regulations and legal frameworks as well as protect and monitor cables using naval and other military resources. International frameworks like the UN Convention on the Law of the Sea (UNCLOS) and the 1884 Convention for the Protection of Submarine Telegraph Cables provide foundational governance for the sector. UNCLOS, for example, has been ratified by 171 parties and helps inform domestic laws and regulations. However, both are outdated and inconsistently enforced. Other organizations promote cable protection best practices, coordinate security responses, and promote building a trusted cable network. This work includes

providing an avenue for coastal states with subsea cable interests to fill in policy and regulatory gaps left by outdated or nonexistent frameworks.

As of July 2025, the United States has 90 cables registered, either in operation or planned. Oversight for these systems is fragmented across federal, state, and tribal agencies, with no lead agency to streamline and facilitate policymaking, permitting, and regulatory processes. Key federal stakeholders include the Federal Communications Commission, which provides licenses and oversees rules and regulations; the Department of Homeland Security, which acts as the national coordinator for critical infrastructure, monitors cable-laying vessels, and develops interagency concepts for cable security and resilience in crises; the Department of State, which engages with global governments and institutions on subsea cable treaties and diplomacy and oversees digital policies and funding for digital infrastructure projects; Team Telecom, which reviews foreign investment in cable licenses; and U.S. federal financing and export credit agencies, which provide feasibility studies, project preparation financing, and project financing for global subsea cable projects.

While frameworks and regulations exist to guide installation of new cables as well as the repair of damaged cables, geopolitical tensions and active conflict are complicating efforts to build in redundancy and resiliency into the infrastructure. The South China Sea and the Red Sea, in particular, have become complicated and have delayed new projects and repairs. China has dragged out approvals or denied new projects for any cables that cross their territorial waters. Other cable projects have been rerouted to avoid China, adding more costs and time needed for construction. Private sector companies, the United States, and like-minded partners and allies are increasingly concerned about China's role in the industry through HMN Tech and other state-directed investments in the infrastructure, as well as China's growing ability to deploy coercive methods such as denying permits.

At the same time, suspected Russian and Chinese activity has demonstrated the considerable risk posed to cable systems from deniable gray zone activities. Multiple incidents, including in the Baltic Sea and the Taiwan Strait, point to the potential weaponization of cable sabotage to disrupt adversaries at minimal cost and with few avenues for recourse. Given the strategic value of subsea cables, such activity, including the use of nominally civilian vessels to sabotage cables, highlights the fraught potential environment for cable security, particularly in the context of any potential NATO-Russia or Taiwan contingency.

Cable breakages in the Red Sea present a different security concern, particularly around the safety of the crew tasked with fixing the cable. Ships passing by the southern coast of Yemen are subject to attacks from Houthi terrorists as well as pirates. The Houthis have launched antiship missiles as part of an effort to air their anti-Israeli sentiments over the Gaza war. In early 2024, a UK-owned ship was hit by multiple missiles and forced to drop its anchor, subsequently damaging cables. Though not a direct attack on cables themselves, this example illustrates the indirect threats to the infrastructure. These risks are likewise increasing the challenges of insuring cable repair ships, with insurance becoming highly challenging to obtain in the region.

Taken together, the risks posed to subsea cables are proliferating in tandem with their critical role to the world's global digital systems and its ever-expanding need for data. Any broader effort to ensure the resiliency and redundancy of subsea cable systems will need to account for this array of challenges, ranging from environmental or incidental damage to malicious sabotage in peacetime or war.

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Table 1: International Policy and Frameworks

Organization	Responsibilities
International Cable Protection Committee (ICPC)	The ICPC promotes the safeguarding of international cables against man-made and natural hazards. It has more than 160 members in more than 60 countries, representing 97 percent of the world's subsea telecom cables, including cable operators, owners, manufacturers, industry service providers, and governments. The ICPC provides recommendations on subsea cable laying and resiliency best practices.
International Telecommunications Union (ITU)	The ITU is a specialized UN agency for digital technologies and is made up of 194 member states and more than 1,000 companies, universities, and international and regional organizations. It convenes multistakeholder dialogues on cable network resiliency, and the ITU, in partnership with the ICPC, has an International Advisory Board on Submarine Cable Resilience to promote dialogue and ways to collaborate to improve subsea cable resilience.
United Nations Convention on the Law of the Sea (UNCLOS)	UNCLOS is an international treaty that governs the world's oceans and activities in and on the seas and defines territorial claims and other jurisdictions for states.
International Seabed Authority (ISA)	The ISA is an autonomous international organization established under UNCLOS. It works with members to govern mineral resources activities, such as deep seabed mining, to protect the marine environment.

Document 4

"Is generative AI a general-purpose technology? Implications for productivity and policy", by Flavio Calvino, Daniel Haerle, Sarah Liu, June 2025, *OECD Artificial Intelligence Papers No. 40*, *www.oecd.org* (extrait)

Executive Summary

The recent rise in generative AI's applications and diffusion have highlighted unprecedented technological advances and mainstreamed some of the technology's growing potential. As businesses and governments focus their efforts on these developments, and the related governance issues, discussions have emerged about generative AI's potentially transformative effects and whether the technology will bring significant benefits in the form of widespread productivity increases and substantially higher living standards and well-being.

Historically, transformational technological innovations like the steam engine, electricity, and computers had profound impacts due to their general-purpose nature, permeating various economic processes and contributing to widespread productivity growth. While previous studies have attempted to evaluate the aggregate productivity effects of AI more broadly, this paper specifically analyses whether *generative* AI is a general-purpose technology, and its broader economic and social implications.

Through a review of theoretical literature and early empirical evidence, including novel descriptive analysis, this study suggests that generative AI has considerable potential to qualify as a new general-purpose technology (GPT). Despite the early stage of technology diffusion upon which the discussion and evidence presented are based, generative AI appears to exhibit defining characteristics of GPTs, notably i) pervasiveness, ii) continuous improvement over time, and iii) innovation spawning. Attaining these three defining GPT characteristics in combination is essential in enabling a considerable economic impact of the technology, as generative AI's potential critically depends on the extent to which it can achieve widespread use that also generates substantial spillover effects and fosters innovational complementarities, raising overall productivity and driving economic growth.

In terms of pervasiveness, the applications of generative AI innovation, despite being prevalent in software, span across a considerable number of other areas, ranging from life and medical sciences to banking and finance. While the most recent official figures on usage among firms appear still limited and concentrated in certain sectors, adoption among individuals appears higher, and the technology displays diffusion potential – based on the tasks it can affect – across diverse sectors that typically account for a considerable share of activities.

Regarding continuous improvement over time, generative AI has benefited from improvements in compute and model capabilities since its inception that point towards a rapidly evolving technology. Ever more capable models are continuously being released, with the most recent ones excelling at various tasks regarding literacy, numeracy and logical reasoning, often even outperforming humans. Additionally, developments in the technology, such as AI agents, are broadening its potential by creating new use cases for the technology.

Considering innovation spawning, the analysis highlights that generative AI patents are relevantly cited by follow-on innovation in a broad number of application areas. Although these notably include, to a large extent, computer technologies and IT methods, they also considerably go beyond those, with applications ranging from medical technologies to logistics. The evidence also uncovers relevant positive feedback loops from follow-on innovation back to generative AI innovation, which suggest the likely relevance of generative AI as invention of a method of invention (IMI). Furthermore, generative AI is changing processes related to research and innovation – likely improving their productivity – and, at the same time, affecting several applied fields, notably including drug discovery and education.

Similarly to earlier GPTs, generative AI may exhibit a productivity paradox, as productivity gains may not materialise immediately and may be dependent on workers' skill development, the implementation of

organisational changes and other complementary investments or innovations. Nonetheless, given the abovementioned time lags and the role of innovational complementarities, the evolution of earlier GPTs seems to provide encouraging signs that generative AI could lead to substantial improvements in productivity in the future, notably through the innovation-spawning channel.

The full realisation of generative AI's productivity potential in the long-term depends however on the implementation of a mix of policies aimed at fostering its widespread diffusion and trustworthy use across firms and sectors, supporting its continuous improvements over time, and encouraging its ability to spawn innovations in application sectors. The three defining characteristics of GPTs appear in fact particularly relevant from a policy perspective, highlighting the relevance of policy levers that not only foster technological adoption and diffusion, but also enable the effective use of generative AI and its applications across sectors alongside relevant organisational changes, and support innovation at the frontier. As the technology advances, ensuring that generative AI's use remains innovative and trustworthy and respects human rights and democratic values, in line with the OECD AI principles, will remain equally critical. These efforts will be instrumental in unlocking generative AI's full transformative potential as a GPT to drive innovation, enhance productivity, and achieve meaningful societal advancements.

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Document 5

"Canada fought measles and measles won; virus now endemic after 1998 elimination", by Beth Mole, 10 November 2025, *Ars Technica,*

www.arstechnica.com

Canada has lost its measles elimination status, meaning the highly infectious virus is considered endemic once again in the country, The Pan American Health Organization (PAHO) announced Monday.

The determination was made by a committee of PAHO experts, who spent last week poring over disease data to assess the measles status of countries across the entire region. The fact that Canada has lost its elimination status means that the region of the Americas overall has also lost the status, which it achieved in 2016. Of the 35 countries and territories in the region—a health region designated by the World Health Organization—Canada is currently the only country where measles is considered to be spreading endemically, though other countries, namely the US and Mexico, are headed in the same direction.

Measles is considered eliminated when a country can go 12 months without continuous local spread. Sporadic cases brought in from international travel can continue to occur, potentially causing limited outbreaks. But elimination is lost and endemicity is declared only when transmission is sustained over the course of a year.

Canada achieved elimination in 1998. The US did the same in 2000. Elimination was achieved through hard-fought vaccination campaigns, as two doses of the measles, mumps, and rubella (MMR) vaccine is 97 percent protective against the virus, and that protection is considered lifelong. But, since that time, vaccine misinformation and potent rhetoric from anti-vaccine activists—including current US Health Secretary Robert F. Kennedy Jr.—have taken hold, driving vaccination rates down on a population level. While the vast majority of American and Canadian parents continue to vaccinate their children, certain pockets and close-knit communities have become dramatically undervaccinated, providing potential footholds for the virus.

That is the case in Canada, where a massive outbreak began in October 2024. It started in Canada's province of New Brunswick, and cases linked to the outbreak have since been detected across the country. As of November 1, the Government of Canada has tallied 5,162 measles cases since the start of the year. In the PAHO meeting last week, experts evaluated genetic data from the cases and determined that the same strain that sparked the outbreak last October has continued to spread, surpassing the 12-month threshold. Although transmission has slowed, the virus is still spreading. There were 23 cases identified in the week of October 26 to November 1.

"This loss represents a setback, of course, but it is also reversible," Jarbas Barbosa, director of PAHO, said in a press briefing Monday.

Call to action

Barbosa was optimistic that Canada could regain its elimination status. He highlighted that such setbacks have happened before. "In 2018 and 2019, Venezuela and Brazil temporarily lost their elimination status following large outbreaks," Barbosa noted. "Thanks to coordinated action by governments, civil society, and regional cooperation, those outbreaks were contained, and the Region of the Americas regained its measles-free status in 2024."

On Monday, the Public Health Agency of Canada released a statement confirming that it received notification from PAHO that it had lost its measles elimination status, while reporting that it is already getting to work on earning it back. "PHAC is collaborating with the PAHO and working with federal,

provincial, territorial, and community partners to implement coordinated actions—focused on improving vaccination coverage, strengthening data sharing, enabling better overall surveillance efforts, and providing evidence-based guidance," the agency said.

However, Canada isn't the only country facing an uphill battle against measles—the most infectious virus known to humankind. Outbreaks and sustained spread are also active in the US and Mexico. To date, the US has documented at least 1,618 measles cases since the start of the year, while Mexico has tallied at least 5,185. Bolivia, Brazil, Paraguay, and Belize also have ongoing outbreaks, PAHO reported.

As of November 7, PAHO has collected reports of 12,593 confirmed measles cases from 10 countries, but approximately 95 percent of them are in Canada, Mexico, and the US. That total is a 30-fold increase compared to 2024, PAHO notes, and the rise has led to at least 28 deaths: 23 in Mexico, three in the United States, and two in Canada.

The PAHO used Canada's loss as a call to action not just in the northern country, but the rest of the region. "Every case we prevent, every outbreak we stop saves lives, protects families, and makes communities healthier," Barbosa said. "Today, rather than lamenting the loss of a regional status, we call on all countries to redouble their efforts to strengthen vaccination rates, surveillance, and timely response to suspected cases—reaching every corner of the Americas. As a Region, we have eliminated measles twice. We can do it a third time."

"PAHO calls for regional action as the Americas lose measles elimination status", 10 November 2025, *Pan American Health Organisation*,

www.paho.org

Washington, D.C., 10 November 2025 (PAHO) –The Pan American Health Organization (PAHO) announced today that the Region of the Americas has lost its verification as free from endemic measles transmission.

The decision follows the conclusion of the PAHO Measles, Rubella, and Congenital Rubella Syndrome Elimination Regional Monitoring and Re-Verification Commission, which met in Mexico City from 4 to 7 November 2025 to review the epidemiological situation across the region. The Director of PAHO, Dr. Jarbas Barbosa, received and agreed with the Commission's report indicating that endemic transmission of measles has been reestablished in Canada, where the virus has circulated for at least 12 months.

As a result, the Americas, which was the first region in the world to eliminate measles twice, has now lost its measles-free status. All other countries continue to maintain their measles elimination status.

"This loss represents a setback—but it is also reversible," said Dr. Barbosa, "Until measles is eliminated worldwide, our Region will continue to face the risk of reintroduction and spread of the virus among unvaccinated or under-vaccinated populations. However, as we have demonstrated before, with political commitment, regional cooperation, and sustained vaccination, the Region can once again interrupt transmission and reclaim this collective achievement."

Current situation

As of 7 November 2025, 12,596 confirmed measles cases have been reported across ten countries (approximately 95% of the Region's cases in Canada, Mexico and US), a 30-fold increase compared to 2024. Twenty-eight deaths have been recorded: 23 in Mexico, 3 in the United States, and 2 in Canada.

Active outbreaks are ongoing in Canada, Mexico, the United States, Bolivia, Brazil, Paraguay, and Belize, mostly triggered by imported cases. Transmission has primarily affected under-vaccinated communities, with 89% of cases occurring in unvaccinated individuals or those with unknown vaccination status. Children under 1 year of age are the most affected, followed by those aged one to four years.

Canada's measles outbreak began in October 2024 in the province of New Brunswick and has since spread nationwide, with more than 5,000 confirmed cases reported nationwide. Although declining, transmission persists in Alberta, British Columbia, Manitoba, and Saskatchewan.

Measles is highly contagious. A single infected person can transmit it to up to 18 others. Severe complications can include pneumonia, encephalitis, blindness, and death. Outbreaks also disrupt daily life and place additional pressure on health systems.

Vaccination remains the most effective means of protection. Over the past 25 years, the measles vaccine has prevented more than six million deaths across the Americas —and an estimated 15 million deaths over the last 50 years. Nevertheless, in 2024, regional coverage for the second dose of the measles, mumps and rubella vaccine (MMR2) averaged 79%—well below the 95% needed to prevent outbreaks. In addition, only 31% of countries reached 95% or more coverage for the first dose, and just 20% achieved that level for the second dose.

PAHO's response and recommendations
PAHO continues to provide technical support to countries to strengthen surveillance, laboratory diagnosis, outbreak response, and vaccination campaigns. Experts have been deployed to Mexico, Argentina, and Bolivia, and the organization is monitoring risks in Belize, Brazil and Paraguay.

Dr. Barbosa emphasized: "Every case we prevent, every outbreak we stop saves lives, protects families, and makes communities healthier. Today, rather than lamenting the loss of a regional status, we call on all countries to redouble their efforts to strengthen vaccination rates, surveillance, and timely response to suspected cases—reaching every corner of the Americas. As a Region, we have eliminated measles twice. We can do it a third time."

RVC strongly encourages countries to:

- Continue strengthening surveillance, immunization and outbreak rapid response activities for every suspect case.
- Expand and consolidate electronic immunization registries across all countries, following PAHO's established guidelines.
- Maintain sustainable laboratory surveillance by allocating sufficient financial and human resources.
- For suspected cases with clinical and epidemiological risk factors and a negative measles IgM result (with specimens collected within 0–3 days after rash onset), perform PCR testing to confirm or rule out infection.
- Implement all recommendations down to the local level.
- Document outbreak closure following the Regional Framework guidelines.

Looking forward
The Americas have previously faced setbacks, including temporary loss of measles elimination status in Venezuela (2018) and Brazil (2019). The region regained elimination status in 2024 through coordinated regional action.

To regain measles elimination status, a country must demonstrate interruption of endemic transmission for at least 12 consecutive months, supported by comprehensive vaccination, surveillance, and outbreak-response data.

Canada will now present and implement an action plan under PAHO's regional framework, focused on boosting immunization coverage, reinforcing surveillance systems, and ensuring rapid outbreak response to stop endemic transmission and regaining measles elimination status.

"Cooperation, solidarity, and science are stronger than any virus—because when we work together, we protect the health and lives of all people in the Americas," Dr. Barbosa concluded.

Document 7

"Kennedy Says He Told C.D.C. to Change Website's Language on Autism and Vaccines",
21 November 2025, by Sheryl Gay Stolberg, The New York Times,

www.nytimes.com (extrait)

Health Secretary Robert F. Kennedy Jr. said in an interview that he personally instructed the Centers for Disease Control and Prevention to abandon its longstanding position that vaccines do not cause autism — a move that underscores his determination to challenge scientific consensus and bend the health department to his will.

In an interview on Thursday explaining why the C.D.C. website now says the claim that vaccines do not cause autism is not "evidence-based," Mr. Kennedy acknowledged that large-scale epidemiological studies of the measles, mumps and rubella vaccine had found no link to autism, and that studies of the mercury-based preservative thimerosal had also shown no link.

But he cited gaps in vaccine safety science. He said he ordered the C.D.C. to change its guidance in part because high-quality large studies had not been conducted to examine a potential link between autism and other shots given in the first year of life. Those include the hepatitis B vaccine and a combination shot that protects against diphtheria, tetanus and pertussis, commonly known as whooping cough.

In his view, sweeping statements like "vaccines don't cause autism" are unproven, and have been deployed by public health leaders who want to ease parents' fears. He said he is not saying vaccines cause autism; he is simply saying there is no proof that they don't.

"The whole thing about 'vaccines have been tested and there's been this determination made,' is just a lie," Mr. Kennedy said in the interview, his first with a major print publication. He added, "The phrase 'Vaccines do not cause autism' is not supported by science."

It is highly unusual for a health secretary to personally order a change to scientific guidance. Former C.D.C. officials say that such changes are usually initiated by agency scientists, and in some cases might go to the secretary's office for review. Mr. Kennedy's intervention flows out of his two decades of vaccine activism, and demonstrates his intense interest in the topic even as other health issues, including the cancellation of insurance subsidies under the Affordable Care Act, are swirling in Washington.

Experts say Mr. Kennedy is correct, in a narrow sense, in saying that certain studies have not yet been done. But they say he is also asking for a level of proof that is difficult to achieve. As Dr. Arthur Caplan, who directs the medical ethics division at New York University, once wryly noted, "You can't prove that Coca-Cola doesn't cause autism either."

Most mainstream public health leaders, including an official at the American Medical Association, a former C.D.C. director and the chairman of the Senate Health Committee, say the question of whether vaccines cause autism has been settled. Mr. Kennedy's critics say he is cherry-picking facts and seeking studies to confirm his own predetermined conclusion.

By focusing on a theoretical, unproven risk, critics say, Mr. Kennedy is using his platform as health secretary to discourage vaccination, an issue that earned him millions of dollars and boosted his political career. They accuse him of playing on parents' fears about autism, while ignoring incontrovertible evidence that vaccines have saved millions of lives.

"There is overwhelming evidence that vaccines do not cause autism but do save lives," said Dr. Mandy Cohen, who led the C.D.C. during the Biden administration. She said the website change damages the

C.D.C.'s credibility and "risks endangering children by driving down vaccination rates and leaving kids vulnerable to preventable diseases like measles and whooping cough."

The website changes have once again put the secretary at odds with Senator Bill Cassidy, Republican of Louisiana and the health committee chairman, who is also a doctor. To win the senator's vote for confirmation, Mr. Kennedy promised not to remove a statement from the C.D.C. website stating that vaccines do not cause autism.

The statement remains at the top of the site, with an asterisk leading to a footnote that says the header "has not been removed due to an agreement" with the health committee chairman.

The page goes on to say that the claim is not "evidence-based" and that "studies supporting a link have been ignored by health authorities." It does not mention the life-saving benefits of vaccines.

Mr. Cassidy, writing Thursday on the social media platform X, appeared incensed.

"I'm a doctor who has seen people die from vaccine-preventable diseases," he wrote. "What parents need to hear right now is vaccines for measles, polio, hepatitis B and other childhood diseases are safe and effective and will not cause autism. Any statement to the contrary is wrong, irresponsible, and actively makes Americans sicker."

Mr. Kennedy said in the interview that he had spoken with the senator about the website changes.

"I did talk to him," Mr. Kennedy said of Mr. Cassidy. "He disagreed with the decision." Mr. Cassidy has recently said he would like Mr. Kennedy to appear before his committee; the secretary said he would be "happy to."

The introduction of routine childhood immunization is widely considered one of the greatest success stories of public health.

A study published last year in the medical journal The Lancet estimated that since 1974, when the World Health Organization launched its global vaccination program, vaccines had averted 154 million deaths worldwide, including those of 146 million children younger than 5. The C.D.C. has cited vaccines as one of the 10 great public health achievements of the 20th century.

Before the pertussis vaccine became widely available in the 1940s, for example, as many as 200,000 people got sick with whooping cough every year in the United States, and about 9,000 died, according to the National Foundation for Infectious Diseases. Before the measles vaccine became available in 1963, about 450 Americans died each year from the disease.

Dr. Sandra Adamson Fryhofer, a trustee of the American Medical Association, said the organization was "deeply concerned that perpetuating misleading claims on vaccines will lead to further confusion, distrust and ultimately, dangerous consequences for individuals and public health."

Concerns about vaccine safety, which have percolated since the 1980s, intensified in 1998, after a British doctor, Andrew Wakefield, published a small study asserting that the measles, mumps and rubella vaccine might be linked to autism. That paper has since been retracted, and numerous studies — including a 10-year study of 600,000 children published in 2019 — have found no association.

A 2018 study that followed 82,000 pregnant women who were given the tetanus, diphtheria and pertussis vaccine found their babies were not at increased risk for autism.

However, Dr. Daniel Salmon, who directs the Institute for Vaccine Safety at the Johns Hopkins Bloomberg School of Public Health, said Mr. Kennedy was partially correct about the limited high-quality, large-scale epidemiological research on autism and infants given the vaccines for hepatitis B and diphtheria, tetanus and pertussis. Dr. Salmon has long advocated greater federal investment in vaccine safety science.

“I think that there are gaps in science that need to be filled,” Dr. Salmon said. “We need a research agenda and we need to make it a priority to answer these questions, but it’s got to be done with rigorous science, and the science is really hard to do.”

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